2022

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Junior
Grades 7–8

TIME ALLOWED

75 minutes

Instructions and Information 考试说明

General 一般说明

1. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 000 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
   试卷包含 25 道单项选择题和 5 道填空题，填空题答案均为 000-999 之间的整数。题目设置由易到难，答错不扣分。
2. NO calculators, maths stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
   考生禁止携带计算器、数学模板尺、手机或其他计算工具，允许使用草稿纸、坐标纸、直尺和圆规。
3. Diagrams are NOT drawn to scale. They are intended only as aids.
   图形只作为辅助参考，并未按照实际比例绘制。
4. This is a competition not a test; do not expect to answer all questions. You are only competing against your own grade in your own country/Australian state so different grades doing the same paper are not compared.
   这是一场比赛，而不是一场测验，你无需在规定时间内做完全部题目。你只会与你所在年级的参赛者进行评比，所以即使做同一份试卷，所在年级不同也不会一同评比。
5. Read the instructions on the answer sheet carefully. Ensure your name, school name, level, school grade and exam code are entered. It is your responsibility to correctly code your answer sheet.
   务必仔细阅读答题卡上的填涂说明。确保准确填写你的姓名、在读学校、考试等级、考试年级和准考证号。
6. When your teacher gives the signal, begin working on the problems.
   监考老师发出作答指令后，方可开始答题。

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder

You may sit this competition once, in one division only, or risk no score.

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The answer sheet 答题卡说明

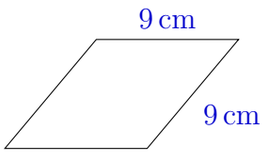
1. Mark your exam code and answers with a **2B** pencil.
   使用 2B 铅笔正确填涂准考证号和答案。
2. Record your answers on the answer sheet provided (not on the question paper) by FULLY colouring the circle matching your answer.
   将答案全部涂在答题卡上（在试卷上作答无效）。
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.
   你的答题卡将通过扫卡机进行扫描，读卡机会读出答题卡上的所有标记，所以不要在答题卡上乱写乱画。修改答案时务必用橡皮擦干净，保持答题卡清洁。

Junior Division

1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问右图菱形的周长是多少？

What is the perimeter of this rhombus?



(A) 20 cm (B) 24 cm (C) 28 cm (D) 32 cm (E) 36 cm

2. 山上的气温为 4 °C，但是经过一晚后下降了 7 °C。
请问早上的气温是多少？

The temperature in the mountains was 4 °C but dropped overnight by 7 °C.
What was the temperature in the morning?

(A) 3 °C (B) 11 °C (C) −3 °C (D) −4 °C (E) −11 °C

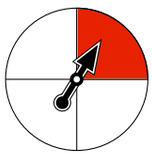
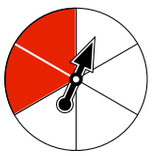
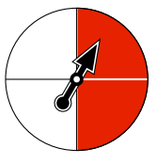
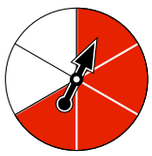
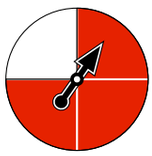
3. 请问算式 20×22 的值是多少？

What is the value of 20×22?

(A) 42 (B) 440 (C) 2022 (D) 2220 (E) 4400

4. 请问哪一个选项中的指针落到红色的可能性是落到白色的两倍？

Which spinner is twice as likely to land on red as white?

(A)  (B)  (C)  (D)  (E) 

5. 罗老师的音乐课从下午 4:28 开始并于下午 5:05 结束。
请问这堂课上了多久？

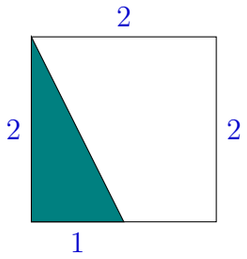
(A) 23 分钟 (B) 27 分钟 (C) 33 分钟 (D) 37 分钟 (E) 43 分钟

Russell's tuba lesson started at 4:28 pm and finished at 5:05 pm.
How long was the lesson?

(A) 23 minutes (B) 27 minutes (C) 33 minutes (D) 37 minutes (E) 43 minutes

6. 请问右图中阴影部分占正方形面积的几分之几？

What fraction of the square is shaded?



(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{2}{5}$ (D) $\frac{1}{2}$ (E) $\frac{3}{8}$

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7. 请问算式 $10^3 - 11^2$ 的值是多少？

What is the value of $10^3 - 11^2$?

(A) 8 (B) 22 (C) 279 (D) 779 (E) 879

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8. 请问下面哪一个分数在数轴上的位置介于 4 与 5 之间？

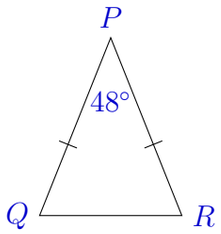
Which one of these fractions lies between 4 and 5 on the number line?

(A) $\frac{7}{2}$ (B) $\frac{15}{4}$ (C) $\frac{16}{5}$ (D) $\frac{17}{4}$ (E) $\frac{18}{5}$

---

9. 如右图所示，在三角形 PQR 中，$PQ = PR$ 且 $\angle QPR = 48°$。
请问$\angle PQR$ 的度数是多少？

In the triangle PQR shown, $PQ = PR$ and $\angle QPR = 48°$.
What is $\angle PQR$?



(A) 60° (B) 66° (C) 72° (D) 78° (E) 84°

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10. 从星期日正午开始再过四分之一个星期是什么时间？

(A) 星期二早上 6 点 (B) 星期二晚上 9 点 (C) 星期一正午
(D) 星期三凌晨 3 点 (E) 星期一下午 6 点

What is the time and day one-quarter of a week after midday on Sunday?

(A) 6 am Tuesday (B) 9 pm Tuesday (C) midday Monday
(D) 3 am Wednesday (E) 6 pm Monday

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 右图三枚硬币的正反两面都各有一个数。已知每枚硬币上的两个数之积都是 60，请问这三枚硬币背面的数之和是多少？

These three coins have a number on each side. The two numbers on each coin multiply to 60. What is the sum of the three hidden numbers?

(A) 17 (B) 21 (C) 29 (D) 31 (E) 39

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12. 如右图所示，三个大小不同的正方形排在一起。已知最大正方形的周长为 36 cm，最小正方形的面积为 9 cm^2。请问中等大小的正方形的周长是多少？

Three different squares are arranged as shown.
The perimeter of the largest square is 36 cm.
The area of the smallest square is 9 cm^2.
What is the perimeter of the medium-sized square?

(A) 12 cm (B) 18 cm (C) 24 cm (D) 30 cm (E) 32 cm

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13. 澳大利亚每天使用 160 百万升汽油。现有储备的汽油足够使用 60 天。请问澳大利亚还需要购买多少汽油才能有足够 90 天的储备量？

(A) 4 百万升 (B) 4.8 百万升 (C) 480 百万升 (D) 160 百万升 (E) 4800 百万升

Australia uses 160 million litres of petrol each day.
There is enough petrol stored to last 60 days.
How much more petrol does Australia need to buy to have enough stored for 90 days?

(A) 4 million litres (B) 4.8 million litres (C) 480 million litres (D) 160 million litres (E) 4800 million litres

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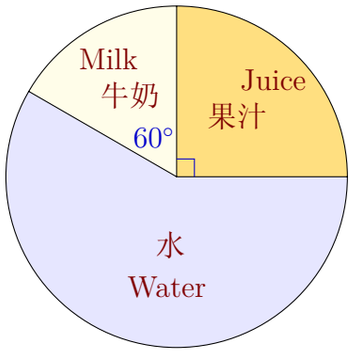
14. 调查学生们最喜欢的饮料：果汁、牛奶或水。右侧的饼图显示了调查结果，有 80 名学生选择牛奶。请问有多少名学生选择果汁？

A number of students were asked about their favourite drink: juice, milk or water. This pie chart shows their answers. Eighty students chose milk.
How many students chose juice?



(A) 80 (B) 100 (C) 120 (D) 160 (E) 480

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15. 请问下面有多少个数字可以被 3 整除？

1, 12, 123, 1234, 12345, 123456, 1234567, 12345678, 123456789

How many of these numbers are divisible by 3?

1, 12, 123, 1234, 12345, 123456, 1234567, 12345678, 123456789

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

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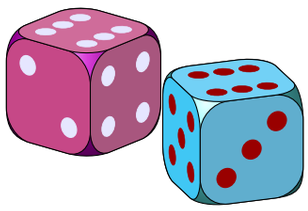
16. 标准骰子上的点数 1 与 6 相对，2 与 5 相对， 3 与 4 相对。调整骰子的放置方式可以看到骰子的 1个、2 个或 3 个面，请问最少要放置多少枚骰子才能恰好一次看到 100 个点？



Standard dice have 1 and 6 opposite, 2 and 5 opposite and 3 and 4 opposite. I can position any dice to be able to see 1, 2 or 3 sides.
What is the smallest number of dice I can arrange to see exactly 100 dots at once?

(A) 5 (B) 6 (C) 7 (D) 10 (E) 17

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17. 将右图所示的标识沿垂直轴 ℓ 翻转，然后顺时针方向旋转 90°，再沿垂直轴 ℓ 再次翻转。请问经过这三步变换后，这个标识是什么样子的？

This logo is reflected in the vertical axis ℓ shown, rotated clockwise by 90°, then reflected in the vertical axis ℓ again.

What does it look like after these three steps?

(A)

(B)

(C)

(D)

(E)

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18. 右侧的益智游戏要求玩家在每个小方格中按照下面两条规则填入 × 或 ○：第一条，每行和每列都恰好包含三个 × 和三个 ○；第二条，每行和每列中都不能有连续的三个相同的符号。

完成这个益智游戏后，请问左下角三个阴影小方格内依次填入的符号是什么？

			○		
×					
	○	○			
			×		×
×		×		○	

In this grid, the puzzle is to fill each square with either × or ○ following two rules. Firstly, in each row and column there must be three of each symbol. Secondly, there can't be three consecutive squares with the same symbol in any row or column.

When this puzzle is solved, what is the arrangement of symbols in the three shaded squares in the lower left?

(A) × ○ ×　　(B) ○ × ○　　(C) × × ○　　(D) ○ × ×　　(E) ○ ○ ×

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19. 小西、小诗和小泰喜爱收集与交换怪物卡片。他们互相见面并交换了一些卡片。小西交换了 11 张卡片、小诗交换了 8 张卡片，小泰交换了15 张卡片。每张卡片都只能与别人的其它卡片交换 1 次。最后，每个人都与最初时拥有的卡片数量相同。请问小诗与小泰互相交换了多少张卡片？

Ash, Sash and Tash like to collect and swap monster trading cards. They meet to trade some cards with each other. Ash trades 11 cards, Sash trades 8 and Tash trades 15.

Each card is swapped exactly once with another card belonging to another person. Everyone ends up with the same number of cards that they started with.

How many cards does Sash swap with Tash?

(A) 1 (B) 3 (C) 4 (D) 6 (E) 7

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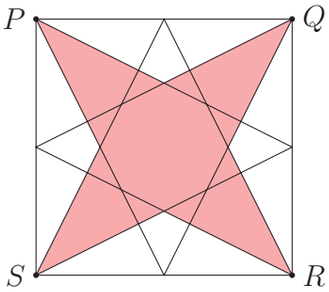
20. 如右图所示，将正方形 $PQRS$ 的每个顶点与其对边的中点连接起来。请问阴影部分占正方形 $PQRS$ 面积的几分之几？

Within the square $PQRS$, lines are drawn from each corner to the middle of the opposite sides as shown. What fraction of $PQRS$ is shaded?

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{3}{8}$ (D) $\frac{1}{2}$ (E) $\frac{2}{3}$



21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 一个数列的首项是 2，之后的每一项都是将前一项各个数位上的数字相加之和再取平方得到的值。请问这个数列的第 2022 项是多少？

The first number in a list is 2. After that, each number is calculated by adding the digits of the previous number together and squaring the result. What is the 2022nd number in the list?

(A) 49 (B) 169 (C) 256 (D) 529 (E) 1024

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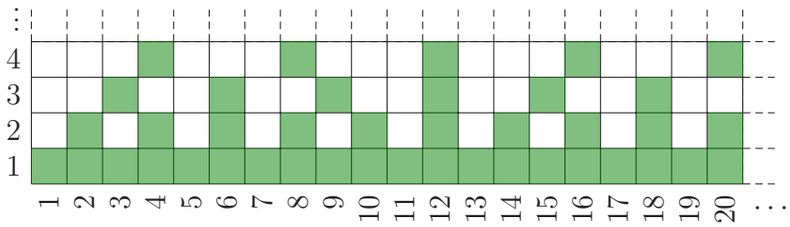
22. 在一个大的正方形网格中，行与列的编号情况如下图所示。
第 1 行的小方格全都涂上阴影；第 2 行的小方格每隔一个涂上阴影；第 3 行的小方格每隔两个涂上阴影；以此类推。
因此每列都有一定数量的小方格被涂上阴影。例如，现在第 6 列有三个小方格被涂上阴影，下一个被涂上阴影的小方格将在第 6 行出现。
请问第 105 列有多少个小方格被涂上阴影？



On a large grid, rows and columns are numbered as shown.
All squares in row 1 are shaded. Every second square in row 2 is shaded. Every third square in row 3 is shaded, and so on.
As a result each column has certain squares shaded. For instance, the shaded squares in column 6 are the three squares shown and one more in row 6.
In column 105, how many squares are shaded?

(A) 3 (B) 4 (C) 8 (D) 12 (E) 35

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23. 通常小安从学校步行回家用时 24 分钟。上周一，他步行了 15 分钟后开始下雨，所以之后他跑步回家。他跑步的速度是平常步行速度的 1.5 倍。请问上周一他从学校到家一共花了多少分钟？

Usually Andrew walks home from school in 24 minutes. Last Monday, he walked for the first 15 minutes but then it started to rain, so he ran the rest of the way home. His running speed is 1.5 times his usual walking speed. How many minutes did it take him to get from school to home?

(A) 18 (B) 20 (C) 21 (D) 22 (E) 23

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24. 一位数的单位分数是 $\frac{1}{2}$、$\frac{1}{3}$、$\frac{1}{4}$、$\frac{1}{5}$、$\frac{1}{6}$、$\frac{1}{7}$、$\frac{1}{8}$ 与 $\frac{1}{9}$。
请问其中有多少对分数满足第一个分数减第二个分数的差大于 $\frac{1}{10}$？

The single-digit unit fractions are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{6}$, $\frac{1}{7}$, $\frac{1}{8}$ and $\frac{1}{9}$.
How many pairs of these fractions are there where the first fraction minus the second fraction is bigger than $\frac{1}{10}$.

(A) 5 (B) 10 (C) 15 (D) 20 (E) 25

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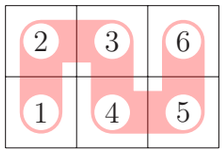
25. 如右图所示，方格表内填入 1 到 6 的数并按照递增顺序相连形成了一条路径。路径可以从一个小方格向相邻的小方格移动形成，但不可以向对角的小方格移动形成。请问要形成这样一条路径，数字 1 到 6 一共有多少种填入方格的方式？



In the grid shown, the numbers 1 to 6 are placed so that when joined in ascending order they make a trail. The trail moves from one square to an adjacent square but does not move diagonally. In how many ways can the numbers 1 to 6 be placed in the grid to give such a trail?

(A) 16 (B) 20 (C) 24 (D) 28 (E) 36

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。
第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, the answer is an integer from 000 to 999.
Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 右图所示的加法竖式中，符号 ♢、♡ 与 □ 分别代表三个互不相同的数字。请问 ♢♡□ 代表的三位数是多少？

$$\begin{array}{r} \diamondsuit\ \heartsuit\ \square \\ +\ \diamondsuit\ \heartsuit\ \square \\ \hline \heartsuit\ \heartsuit\ \diamondsuit \end{array}$$

In the sum shown, the symbols ♢, ♡ and □ represent three different digits. What is the three-digit number represented by ♢♡□?

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27. 将数字 1、2、3、4、5、6、7、8 分为两组，每组 4 个数。将每组数字构成一个四位数然后将这两个四位数相加。最后，将相加之和的各个数位上的数字相加。
例如，$3541 + 7628 = 11169$，其各个数位上的数字之和等于 $1 + 1 + 1 + 6 + 9 = 18$。
请问各个数位上的数字相加之和的最大可能值与最小可能值之差是多少？

The digits 1, 2, 3, 4, 5, 6, 7, 8 are separated into two groups of 4 each. Each group is formed into a four-digit number and the two numbers are added. Finally, the digits in this sum are added together.
An example of this is $3541 + 7628 = 11169$, with digit sum $1 + 1 + 1 + 6 + 9 = 18$.
What is the difference between the largest possible and smallest possible digit sums?

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28. 冲浪俱乐部有三种类型的成员：学徒、教练和专家。其中有 20 名学徒，人数少于成员总数的一半。专家人数是教练的两倍。

一次海滩救援后，俱乐部收到 \$1000 的捐款并全部分给所有成员，每位成员分到的金额都是整数且至少为 \$2 。

每位教练分到的金额是每位学徒的 6 倍；每位专家比每位教练多收到 \$5 。
请问这个冲浪俱乐部一共有多少名成员？

A surf club consists of three types of members: trainees, paddlers and legends. There are 20 trainees, which is less than half the membership. There are twice as many legends as paddlers.

After a surf rescue they received a \$1000 donation to be divided among the members. All the donation was shared and every member received a whole number of dollars, at least \$2.

Each paddler received 6 times as much as each trainee. Each legend received \$5 more than each paddler.

How many members are in the surf club?

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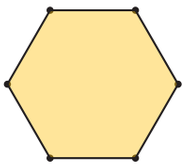
29. 小何有一个面积为 60 的正六边形。他每次选出这个正六边形中的三个顶点，然后写下这三个顶点构成的三角形的面积。
小何一共写下 20 个三角形的面积，请问它们的面积之和多少？



Horton has a regular hexagon of area 60. For each choice of three vertices of the hexagon, he writes down the area of the triangle with these three vertices.

What is the sum of the 20 areas that Horton writes down?

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30. 请问一共有多少种不同的方式将 100 写成三个不同的正整数之和？注意：我们不考虑各项的相加顺序，所以 $34+5+61$ 与 $61+34+5$ 视作同一种方式。

In how many ways can 100 be written as the sum of three different positive integers? Note that we do not consider sums formed by reordering the terms to be different, so that $34+5+61$ and $61+34+5$ are treated as the same sum.